

ENGS 21 Written Report

Spring 2026

Bigger Boulder Better - On Belay

Team 11:

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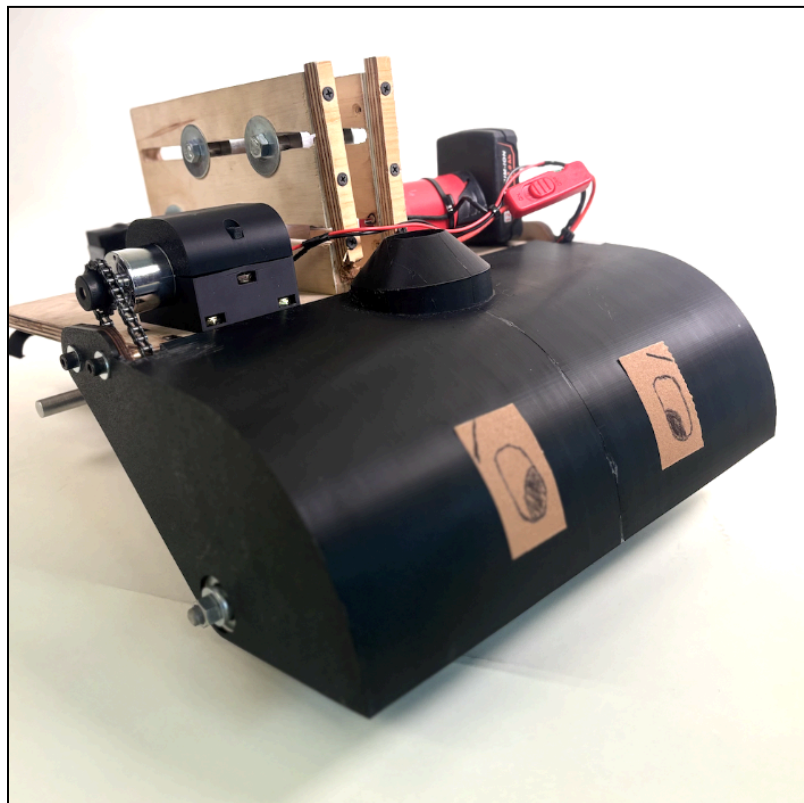


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Problem and User

Indoor Climbing Gyms are difficult to clean, resulting in infrequent cleaning and allowing the buildup of bacteria that can lead to infections.

In 2021, over 5.6 million people visited climbing gyms. With 695 commercial climbing gyms in the US, there are many people that can benefit from our product (Climbing Business Journal). Two climbers on our team have suffered from climbing-wall-born-infections from hobby climbing. Climbing gym walls have been described as “fecal veneer” by researchers (Brauer et al., 2014). Gyms are incentivized to keep a sanitary facility as healthy climbers are ones that will keep coming back. Private and international climbing gyms offer an additional market, though one that could be more difficult to break into.

Our users are climbing gym employees. The product will be purchased by climbing gym owners. The primary beneficiary will be clients, though this benefit to clients will also monetarily benefit gym owners due to marketability of a sanitary facility. Climbing gym employees will also benefit from the streamlined cleaning process due to less labour.

State of the Art

The current state of the art for climbing wall cleaning is the removal of all climbing holds, followed by the removal of all wall panels, the transportation of these panels to an alternate location, powerwashing of the panels, and then the reinstallation of the panels and holds. This is a process that can easily take all day or longer and is usually only done once a year since it requires closing the gym and a lot of manual effort.

Another option for climbing gyms is to belay an individual up the wall with a soap and/or vinegar solution with a rag to clean manually. This process is also highly time intensive and the scrubbing of massive panels (some climbing gym panels can be upwards of 100sqft) across an entire gym is a major undertaking.

Since these processes are major undertakings, climbing gyms avoid cleaning their walls, with many climbing gyms only cleaning once a year. This results in the massive bacteria growth which we identified as our target problem.

Automated wall cleaning technology has been explored in other areas. Window cleaning robots and pool cleaning robots have been patented. These devices, however, do not work for climbing gyms for various reasons. Patent application no. 18/562,897, a window cleaning device we looked at, utilizes a massive liquid spraying system which would create a mess indoors. Patent no. US 9689170 B1 utilizes a large fan to push the device into contact with the surface to be cleaned. This wall cleaning robot would not be suitable for indoor use due to the size of the fan system, the noise, and the dust, chalk, and debris that the fans would put into the air.

While these existing devices cannot be directly used in indoor climbing gym wall sanitation, they served as inspirations and guides for our project.

Specifications and Problem Solving Method

Specification	Justification	Quantification
Spec 1: Ease of Use	If it is not easy to use then it will not be used frequently and won't clean the walls.	Takes less than 15 minutes to set up and run per wall, and should be easily run by a single person.
Spec 2: Efficacy	If device does not remove harmful bacteria and residue then climbers are still at risk.	Kills >80% of germs and removes visible chalk build-up.
Spec 3: Damage-Free	If device damages the climbing wall, gyms will not choose to use it.	No visible damage caused by device.
Spec 4: Speed	If it takes too long to clean then it won't be used.	Entire wall cleaning should take less than 15 minutes.
Spec 5: Durability	If it is not durable, then it won't be able to continue cleaning the walls.	Should be able to clean 50+ walls over lifetime.
Spec 6: Cost	If cost is prohibitive then it won't be used.	Entire system should be <\$300.
Spec 7: Safe	If it is not safe then it won't be used.	Should meet all safety standards for disinfectants, ventilation (chalk dust), and electronics. Users should feel safe using.
Spec 8: Legal	If it is illegal then it cannot be used.	Should meet all US legal requirements.
Spec 9: Ethical	If there are ethical concerns then people are at risk and it won't be used.	Should not cause any ethical concerns.

Our Teams Initial Specifications

Specification Elaboration:

Ease of Use:

Current cleaning solutions are too time and labor intensive to be practical for climbing gyms. The goal is for one person to set up and clean a panel in under 15 minutes with minimal effort. This would make nightly panel cleaning viable, either all at once or on a rotating schedule.

Efficacy:

The device aims to both clean visible chalk and grime and sanitize the wall, targeting at least an 80% reduction in bacteria. Effectiveness will be tested by swabbing and culturing wall samples before and after cleaning with both the current standard method and the new device. This comparison will determine how much the device improves overall cleanliness.

Damage-Free:

Any risk of wall damage would deter climbing gyms from adopting the product. The device must avoid damage whether from misuse, device failure, or normal operation. Walls will be visually inspected after each use to confirm no damage has occurred.

Speed:

While ease of use addresses the labor problem, speed targets the time-intensive nature of current cleaning methods. The device should clean a full wall section — up to 360 square feet — within 15 minutes of hands-on employee time. This distinguishes speed as a separate but closely related specification to ease of use.

Durability:

The device must withstand repeated use without constant replacement, with a target lifespan of 50+ cleaning sessions. Stress testing will substitute for real-world longevity testing where full testing isn't feasible. The device should also be resilient enough to handle improper use without breaking down.

Cost:

The device must be economically viable for climbing gyms to justify purchasing it. A price point above what gyms consider reasonable will kill adoption regardless of performance. The target maximum cost for the full system is \$300.

Safety, Legality & Ethicality:

The device must be safe to operate, legal to own and use, and ethical in its application. These standards apply equally to the testing process, including obtaining proper permissions from local gyms. Compliance with all three is a non-negotiable baseline for the product.

Potential Solutions

Clean-O-Drapes:

The cleaning drapes would be a set of curtains with microfiber towels that would be attached to a rail along the top of the climbing wall. The towels could be continuously soaked with a disinfectant, then pulled against the wall and moved back and forth all the way along the wall to clean the wall. This system would be effective at both cleaning visible residue and disinfecting, and would be a relatively passive system. There would likely be a fair amount of upfront investment from gyms in order to put a rail system above all of the walls, and storing the curtain system when not in use (and properly maintaining it by replacing the microfiber cloths) could be fairly difficult and expensive.

On Belay:

On Belay? Would be a device with two oscillating brushes to clean off chalk residue, and a UV light to kill harmful bacteria. The device would be clipped into existing knots in climbing ropes that are attached to a hard anchor at the top of the wall. The device would be raised and then lowered down the wall as the brushes scrub and UV light disinfects the wall. This system would utilize existing infrastructure in the gym and be very easy for a single employee to operate, but it may take a little longer to clean each region of the gym, so the brushes would have to be wide enough to cover a whole route area.

Wallba:

The Wallba would be a suction powered device that adheres to the climbing wall and uses wheels to navigate the surface of the wall. It would have a vacuum component to remove chalk residue from the wall, as well as a component to spray disinfectant as it travels. This solution would be the most technically challenging to fabricate, as getting it to adhere to the wall would be an extremely difficult task from the get-go. Additionally, there is risk of the device damaging the textured surface of the climbing wall as it travels. The upsides of this solution would include that it could be run while the gym is closed, effectively resulting in no-hands on time for employees, and the potential to easily navigate the different contours of the wall if it is small enough.

Clean-Stik:

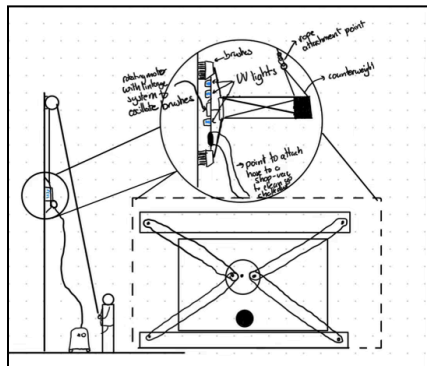
The Clean Stik would be a device that telescopes to reach even the tallest of walls. A scrubber and disinfectant sprayer could be attached to the end of the stick to allow employees to remove both chalk residue and harmful germs. This design is the most similar to the current state of the art, but it allows employees to stay on the ground as they clean the wall. The scrubber attachment could also be made larger to cover a greater portion of the wall, but that risks making the device unwieldy to use at the top portions of the wall. This would be the simplest of all the solutions.

Alternatives Matrix

Solution(1-5 points)	Ease of Use	Sanitization	Damage free	Speed	Durability	Cost	Safe?	Legal?	Ethical ?	Total:
Weight(1-3x)	3x	3x	3x	2x	2x	1x	Yes/No	Yes/No	Yes/No	-
CleanDrapes	3	4	4.5	4	4	5	Yes	Yes	Yes	55.5
On Belay	4	4.5	5	3	3.5	3.5	Yes	Yes	Yes	57
Wallba	5	4.5	2.5	5(timefree)	2	1	Yes	Yes	Yes	46
Clean Stik	2	4	5	1	5	5	Yes	Yes	Yes	47

We weighted ease of use, sanitation and cleaning ability, and damage-free by 3 to represent our most vital specifications. Ease of use and sanitation/cleaning ability are the main driving factors for the creation of our device. When comparing alternatives, a damage-free alternative also was very important to our group since it would both be easier to convince climbing gyms to allow us to test our product and to purchase our product if there was no risk to their facility. Climbing walls are typically a specifically textured surface with a thin plastic coating over a wooden backing, which can make proper cleaning difficult.

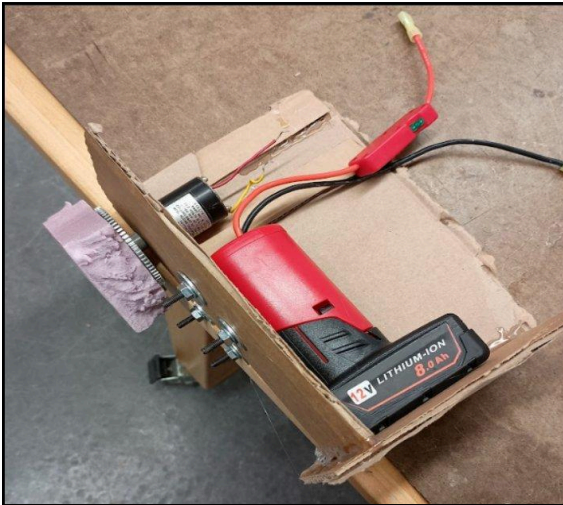
Speed and durability were weighted by 2. Speed is very important to the project but due to its similarity with ease of use, we chose to weigh it below ease of use, though its importance still prompted us to weigh it. Durability was also a metric which our group believed to be important since an easily broken device can be costly both to the facility and to us during testing. Cost was not weighted since our team was fairly confident that a commercial climbing gym would have the budget to purchase any of the devices. This metric was still important for our decision making process, however. We came to our numbers by group vote, each of us provided scenarios and diagrams which demonstrated the ability of the device to adhere to certain specifications. Since we did not have time to test, this was our best method of determining the best device to make: the belay cleaning device. Our backup will be the cleaning drape. The main factor which brought us to the belayed device was our belief that it would be more effective and easier to use since it would actively scrub areas at a speed determined by the user. Furthermore, it would require less upfront investment by the climbing gym than the Clean-O-Drapes, which would require a rail system be installed.



Design Selection:

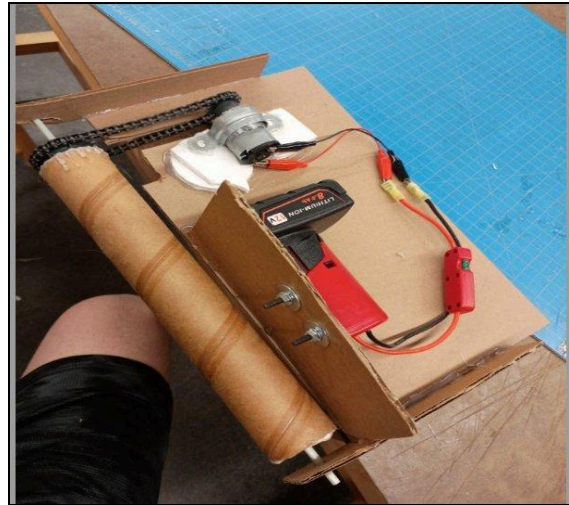
The preceding diagram is of the first sketch of On Belay. We originally imagined the device using oscillating brushes and uv light to clean. A pulley system would bring it up and down and a shop vacuum would clean up chalk. At this point in the design process we also examined options to clean climbing holds as the device belayed down, but decided against it since operating the device during the period where holds are being changed would be excellent timing for cleaning.

Prototyping



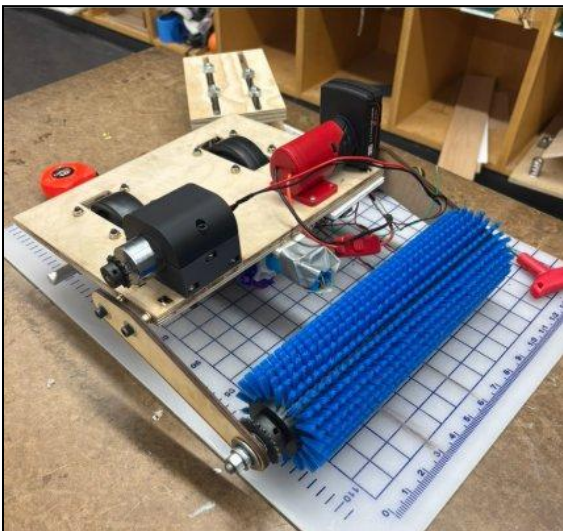
First Prototype

Our first prototype was a simple layout of motor and oscillating brush. After examining this model we realized an oscillating brush was not optimal and began work on our second prototype.



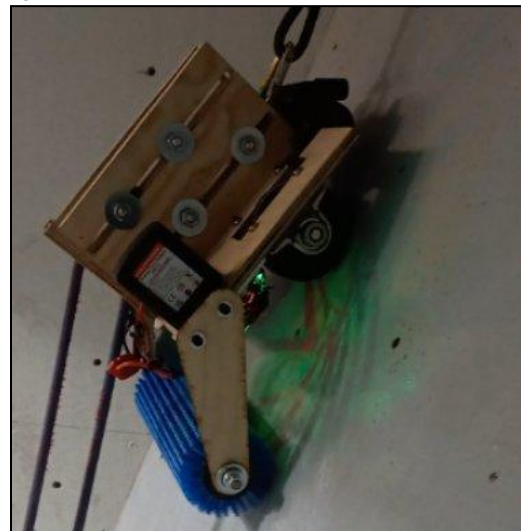
Second Prototype

The second prototype utilized a long spinning brush capable of cleaning the entire wall surface at once. It used the same motor and drill battery but added a chain-driven roller mounted on a fiberglass axle.



Third Prototype

The third prototype established our final body design with a laser-cut chassis, laser-cut brush arms, wheels, and the same motor layout. This served as our looks-like prototype.



Fourth Prototype

The fourth prototype integrated the belaying system and switched from one to two ropes after testing. Brush arms were refined through CAD and trial-and-error. This served as our works-like prototype.

Final prototype/current prototype

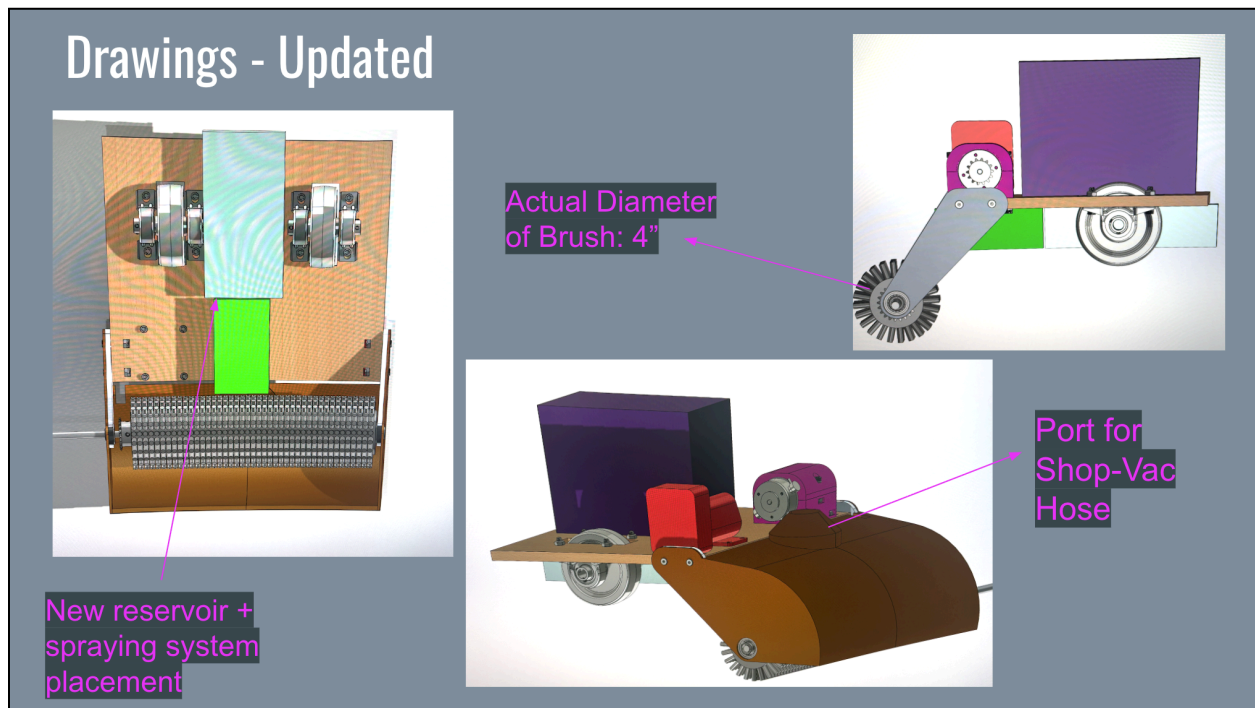
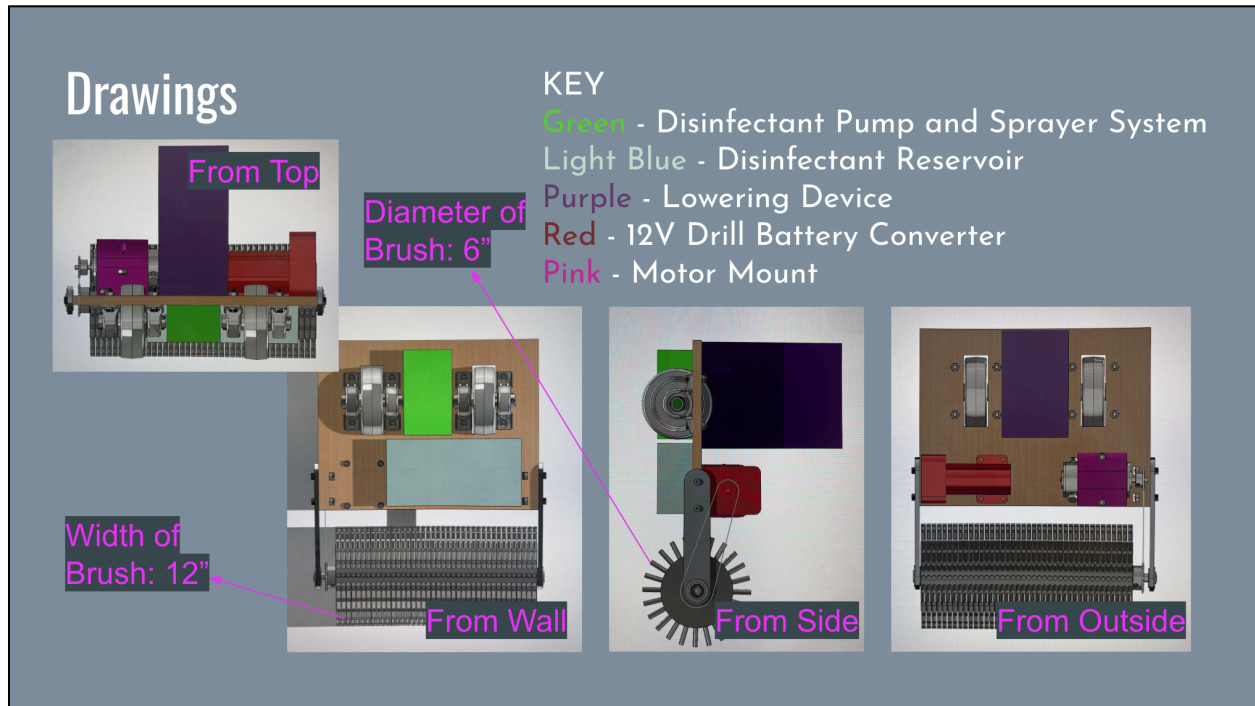
Our 5th prototype was the result of iterating user feedback and attaching the disinfectant-spraying device. Users requested a cover for the brush to prevent it from spraying chalk everywhere, and the spraying system was put together. Users requested that the spraying system was better timed so that it would not make a mess before the device was in position, so a relay was employed that could be customized to prevent the creation of a mess prior to the device being put into position.

Our final prototype consists of 4 subsystems: mechanical (roller & belay), electrical, and fluid. Our device allows a user to thread rope through our belay system and belay it up the wall. At this point, our system will activate the roller and fluid spraying module. As the device is belayed down the wall by the operator, the brush will clean all dust and the spray module will disinfect.

Final Prototype (Photography):



Drawings:



These drawings served to guide our layout as the development of our device progressed. When integrating our separate systems, these drawings proved vital to understanding how these systems would go together. From these drawings we also created 3d printed covers and parts as well as laser-cut parts which made the final product fit together perfectly.

Parts Iteration:



(left) 1 rope system. (right) 2 rope system.

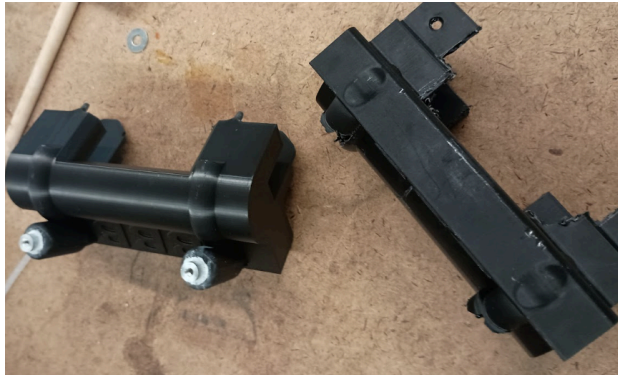
During testing it became clear that a 1 rope system, though easier to use, could not properly tension the device to the wall. This led us to create the two rope system which is present in the final prototype.



(left) various brush arms. (right) incorrect brush arm.

Initially beginning with wooden arms, we transitioned to metal arms, hoping to improve durability. The wood was laser cut and the metal machined. During testing these arms failed to

properly angle the brush to be in contact with the wall. Following that, through trial and error, we made our way through many different arms until we found the perfect ones for our system. We did not make new metal arms once we found the proper shape and size due to time constraints.



Spraying mechanisms

We designed and 3d printed spraying mechanisms from which disinfectant could be misted. We printed 2 different attachments, both of which were not quite watertight. Out of time, we mounted the attachments and simply taped the existing spray nozzles to the side of the attachment. In the future we hope to fully integrate these attachments.

The following chart displays the development of our spraying mechanisms:



Parts iteration proved vital for the success of our device. Due to our device being made up of different systems, each one had its own problems and areas of improvement. By solving issues with each individual device, we were able to create a product that was fully functional.

Testing and Feedback

These images display our device in use by three of our group members and one of our users, a volunteer from the climbing gym who happened to be present during one of our testing sessions.



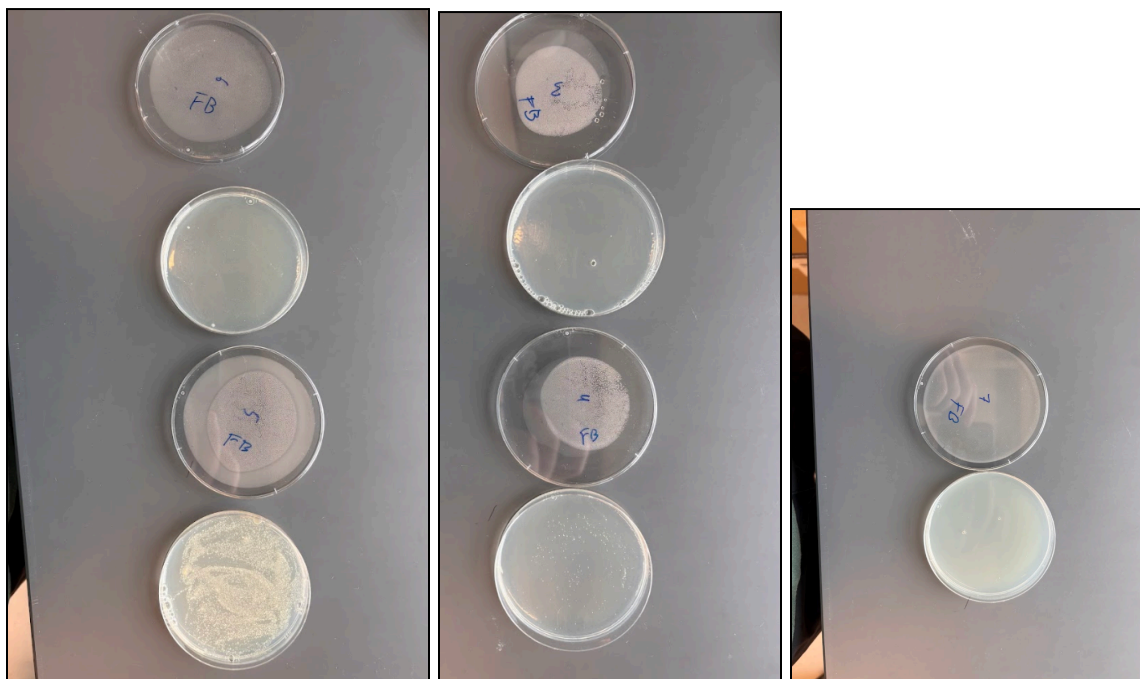
Spec	Target Value	Benchmark Testing	Justification & Testing
Ease of Use	Usable by one person to clean 1ft wide strip of a 30ft (30sqft) tall climbing wall from ceiling to floor.	Takes more than 1 person to remove and clean wall panels.	Will be tested by belaying device down various sections of Dartmouth/The Nook Climbing gyms. Will be tested on various boulders in gyms. Must be easier to use than existing solutions to be marketable.
Efficacy	Kills >80% of germs	Results of test can be found below.	Tested using swab test. Must actually clean to be marketable.
Speed	Takes less than 15 minutes to run device on 1 wall section (30ft tall, 1ft wide (30sqft)).	No specific benchmark testing done due to difficulty and time consumption. Reported time: all day.	Will be tested using stop watch on multiple runs, average taken. Must be considerably faster to be marketable.
Durability	Device can survive a drop of 15ft.	Existing solutions face no durability concerns.	Must be durable enough to be used multiple times. We will stress test durability by dropping it from heights to simulate something going wrong.
Damage-Free	Device does not alter wall.	Existing solutions do not damage climbing walls.	The device cannot damage the wall or it will not be used. Will use human observation since quantification techniques such as microscopy not feasible.
Price	Less than \$300 to make	A power washing rental costs \$50 for a day, workers are paid \$15 dollars an hour.	If device is too expensive it will not be used. Will aim to be less than \$300. Can use surveys to determine if in price range of gyms.
Safety	No slips or slides during tests.	Existing solutions are safe.	In order to be used the system must be safe. We will test safety by taking note of any slips during testing. Will use pulley system that generates enough friction to hold device.
Usability	Can handle 40 degree wall angles.	Existing solutions can handle all angles of wall since state of the art is to remove walls.	Will be tested by belaying device down various sections of Dartmouth/The Nook Climbing gyms. Will be tested on various boulders in gyms. Must be able to handle 40 degree wall angles in the gym.
Speed of setup	Takes less than 5 minutes to set up	Existing setup for wall cleaning requires removing walls and is an all-day procedure.	The actual cleaning of the wall is not the only aspect of speed we will improve. Will time setup of device on wall section from the point of device in area to ready to clean.

Our refined specifications.

As our prototyping process continued and On Belay took shape, new specifications had to be created which were more tailored to our device. Ease of use became a measure of how many people it would require to use the device, speed became a specific measure of sqft/min, setup time was taken into account, safety was tailored to measure a metric relevant to the use of our device, durability was also tailored to the new device as a drop test, and usability was created to determine the device's ability to clean different areas of the climbing gym. Price and efficacy stayed the same as these were targets that still fit our device.

Our benchmarks for state of the art were largely absent due to the lack of a state of the art capable of doing what we sought to do. Speed was benchmarked as all day, ease of use as more than one person, price, and efficacy were all able to be benchmarked, but our other metrics were not present in the state of the art.

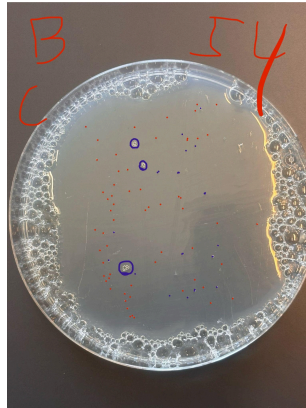
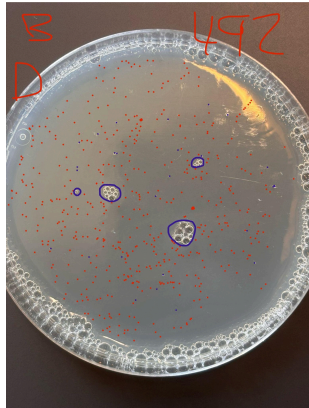
For benchmark testing we swabbed the climbing gym walls to determine bacteria count alongside controls. The following images display our results.



Agar Plates

Wall pre-cleaning: 4 | Wall post-cleaning: 3 | Positive control: 6&7 | Negative Control: 7

This benchmark test measured both the wall without cleaning and with existing cleaning methods alongside a negative and positive control (kitchen sink and no swab) to create an effective measure for our later testing.



Post-cleaning bacteria

Our efficacy test yielded a 89.63% reduction in bacteria. This beat our benchmark of 80% and was comparable to the state of the art.



Damage to wall

Our device did not damage the wall, meeting our set benchmark standard as well as the state of the art.



Ease of use

Our device is usable by one individual, meeting our set benchmark standard and beating the state of the art.



Durability

Our device did not meet the 15ft drop benchmark standard for durability. Our device survived a 5ft drop onto mats. The device did, however, survive during standard use. State of the art does not face durability concerns.



Time Spent

Our device beat the 30sqft in 15min and less than 5 min setup benchmark, taking only 5 min and 45 sec for a 30sqft wall and this included setup time. This also beat the state of the art of an all day process.



Usability

Our device did not meet the usability benchmark of having the capability to operate on 40 degree inclines. Our device was only usable on positively inclined surfaces, flat surfaces, and up to 5 degree angled surfaces. The state of the art has no usability concerns.

Microcontroller	\$4.00
Step Down Breakout	\$4.33
Relay Breakout	\$3.50
Motor	\$14.99
Brush*	\$88.49
Sprockets	\$38.25
Bearings	\$43.80
Screws	\$29.80
Wheels	\$6.70
Swiffer*	\$26.94
Total	\$260.80

Price

Our device beat the \$300 benchmark, costing only \$260.80 to make. The state of the art is a powerwashing rental at \$50 for a day and labour being about \$15 per hour.



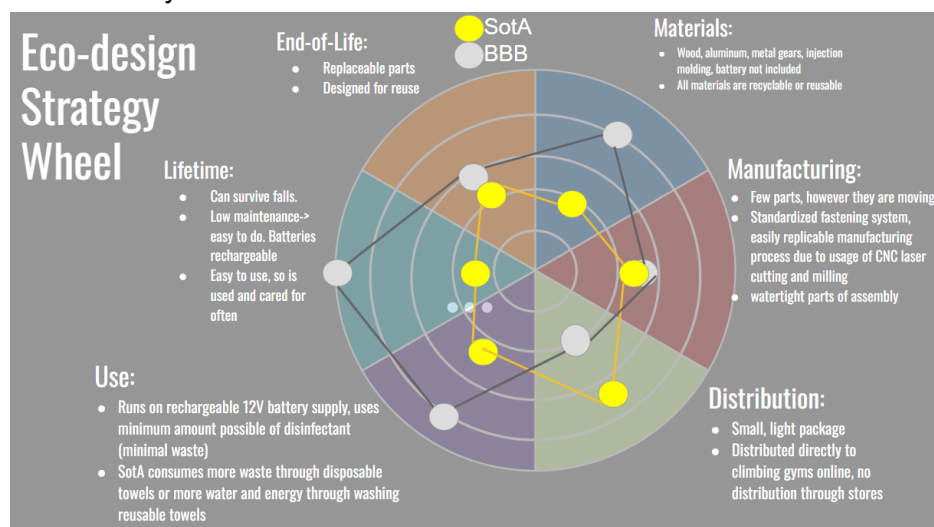
Safety

Our device met the benchmark standard of no slips or drops during testing. The state of the art has no safety concerns.

Feedback from users was taken into account and used to iterate prototypes. Users commented on the exposed chains, brush, and electronics that appeared dangerous. We created 3d printed covers for these parts to protect both the parts and users. Users commented on the inconsistency of our system due to the chain jumping off. By adjusting the brush angle, we were able to keep the chain tensioned, preventing this jumping. Users commented on the unintuitive nature of the use of our device. We improved instructions to users and simplified the two rope system, but this is an area that could use additional work. Turning the brush and sprayer system on before moving the device up the wall created a mess and users found it to be difficult to raise it because of this. We added a relay system to resolve this, though continued development is necessary to find a solution that does not require additional programming for each device sold.

Sustainability and Ethics

Sustainability:



Ethics:

Initial Reactions	
Do you foresee any problems with the proposed solution, product, or process?	Individuals who are not familiar with the belay process could drop the device on themselves. Failure of infrastructure is also a concern, though local gyms must upkeep this infrastructure for their climbers already.
Any safety concerns?	The device could drop from a significant height on a person.
Materials	
How many different materials are required to make the product?	3 - wood, metal, plastic
Are any of the materials toxic? flammable? dangerous?	Flammable. Battery contained, could be toxic if damaged.
What will happen to the product once it wears out or is no longer usable?	Parts can be replaced. Battery can be recharged. Reservoir can be refilled.
Are materials recyclable? Are materials minimized?	Wood, metal, and battery are all recyclable. Plastic elements can be more difficult to be recycled, but can be.
Manufacturing	
How and where will the product be manufactured?	Dartmouth college by our team in college facilities.
Will the use of materials for this product cause shortages for other products or uses?	No, all materials are common. Demand is not widespread-niche.
Transportation and Packaging	
How will the product be transported and distributed?	Ordered online and shipped using UPS.
How will the product be packaged?	Cardboard box and corrugated cardboard. Will use recyclable packing nuggets to prevent damage during shipping.
Unintended Uses	
Do you foresee any unintended uses of the product?	No. Since the device can only be lifted using a belay system, climbing gyms are the only place where the infrastructure exists to even attempt to use device.
Is operation of the product intuitive?	For individuals experienced with belay systems, yes.
What would happen if a child were to use the product?	Child could misuse belay system resulting in the device falling. If a person was underneath it could fall on that person. The child could drink from the cleaning solution reservoir.
Could anyone be harmed during testing?	If product falls on an individual or cleaning fluid enters eyes/mouth injury could occur.
Inclusivity	
Will the product be available to all? Is the cost reasonable for all?	This product will be available to all, though it will only be marketed at climbing gyms. The cost will be reasonable for climbing gyms.
Other	
Do you foresee any other issues or concerns?	None

The climbing wall cleaner addresses ethical concerns primarily around safety - the device could fall from height onto a person or expose users to cleaning chemicals - mitigated by restricting operation to belay-certified staff. On sustainability, the eco-design strategy wheel shows strong performance across most dimensions: all materials (wood, aluminum, metal) are recyclable, manufacturing uses CNC processes with minimal parts, the device runs on a rechargeable 12V battery with minimal disinfectant waste, and it is designed for longevity with replaceable parts and reuse at end-of-life. The main sustainability weaknesses are the battery and plastic injection-molded components, which could be improved through certified-recyclable sourcing or material substitution.

Business Plan Canvas:

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
Climbing equipment manufacturers	Cleaning more effectively, more efficiently	Effective Sanitation Less labor	Forming a working relationship with climbing gyms so that they can seek support with operating the device and so they can also seek replacement parts or maintenance on the device	Climbing gym managers and owners Climbing gym employees Home gym owners
Dartmouth Climbing Gym	Making the device low profile, unobstructive and easy to use	Reduced Time Easy to maintain (uses parts compatible with batteries and vacuums the gym already has)		
Commercial Climbing Gyms	Key Resources Product manufacturers Material distributors Rentable Makerspace/workhouse		Channels Distribution channels may include forming a relationship with large climbing gym chains in order to make the product more widespread We can also travel to demo devices at climbing gyms	
Cost Structure		Revenue Streams		
One-time cost for purchasing of device.		Product Cleaning solution refills		

• Can sell cleaning solution refills in proper container, or in bulk • Can sell replacement parts for brush and other key components (brush cover).	Spare parts
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Economic Evaluation

Variable Costs: \$260.80* × Units Produced

Variable Cost Item	Cost/Unit
Microcontroller	\$4.00
Step Down Breakout	\$4.33
Relay Breakout	\$3.50
Motor	\$14.99
Brush	\$88.49
Sprockets	\$38.25
Bearings	\$43.80
Screws	\$29.80
Wheels	\$6.70
Swiffer	\$26.94
TOTAL	\$260.80*

**Mass production price lower; parts at full scale may cost more.*

Fixed Costs: \$134,000 per year

Fixed Cost Item	Amount
Rent (Manufacturing & Warehouse)	\$5,000/mo
Travel Costs (Gym Demos)	\$2,000/yr
Insurance / Legal	\$7,000/yr
Owner Salaries (×4)	\$120,000/yr
TOTAL	\$134,000/yr

Breakeven Volumes

Breakeven Metric	Value / Notes
Selling price per unit	\$600
Mass production cost per unit	\$200
Profit per unit (device only)	\$400
Units to break even (device only)	335 units/yr
Commercial climbing gyms in US	695
Assessment (device only)	~48% market share required
Refill revenue per device/year	~\$200/yr
Effective profit (with refills)	\$600/unit
Units to break even (with refills)	223 units/yr
Assessment (with refills)	More sustainable; innovation needed

As-is, our business model is not sustainable. This is due to the fact that, selling at our break-even volume, we would be out of clients in about 3 years, as there are only 695 commercial climbing gyms in the US. However, by

expanding markets to private climbing gyms and international gyms, our business model could still be feasible, especially if we continue to generate new innovative products.

Improvements and Takeaways

The team identified several areas for improvement in their climbing wall cleaner, including enabling the device to handle more overhung walls, increasing durability by replacing wood and 3D-printed parts with metal and more robust plastics, adding automatic on/off functionality, and widening the cleaning surface. User feedback echoed these priorities, flagging the device as too heavy, the two-rope system as unintuitive, and calling for a lower profile and automatic lowering/raising. Key struggles included managing liquid in the gym environment, deciding between a one- vs. two-rope system, scale model weight and cost concerns, and spray module reliability. From these challenges, the team drew important lessons: Greta emphasized testing early and often rather than relying solely on CAD; Jonah stressed designing for modularity to allow quick part swapping; Finn highlighted the importance of making all components removable and iterable; and Gaven noted that testing individual modules is not the same as testing an integrated device, and that getting parts to work together proved to be the most critical challenge of all.

References

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